

**FIELD CHECKLIST**

Steam Turbine Overhaul Checklist

Step-by-step field checklist for major and minor steam turbine overhauls — covers clearances, balancing, torque sign-off, and pre-commissioning checks.

Reference standards: API 611 / API 612 (general- and special-purpose steam turbines), API RP 687 (rotor inspection practice), ISO 21940-11 (rotor balance quality grades), ISO 20816-3 (vibration severity evaluation), and ISO 4406 (lube oil cleanliness classification). This checklist summarizes common practice under these standards for field use.

CUSTOMER / SITE		TURBINE MAKE / MODEL	
TURBINE RATING (kW)		JOB / WORK ORDER NO.	
OVERHAUL TYPE	☐ Major ☐ Minor ☐ Emergency	DATE STARTED	

1 Pre-Shutdown Preparation

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Review last overhaul report and outstanding defect list	Previous OH record	☐	
2	Confirm spares, gaskets, and consumables kitted	Per BOM / RFQ	☐	
3	Isolate steam, drain, and lock out lube oil system	LOTO procedure	☐	
4	Record baseline vibration, bearing temps, and running clearances	Log sheet attached	☐	

2 Casing & Rotor Dismantling

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Match-mark all casing joints, pipes, and couplings before removal	Paint / stamp marks	☐	
2	Inspect casing joint faces for erosion / distortion	Visual + DP test	☐	
3	Lift rotor using certified sling; support on V-blocks	Rated capacity sling	☐	



#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
4	Check rotor run-out (bow) at coupling and mid-span	Max 0.05 mm TIR	☐	
5	Inspect blades, shrouds, and lacing wires for erosion/cracks	OEM wear limit chart	☐	

3 Bearings, Seals & Clearances

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Measure journal bearing diametral clearance (top & side)	Typ. 0.001-0.0015 x shaft dia.	☐	
2	Inspect and record babbit condition (wire, wipe, pitting)	Photograph if damaged	☐	
3	Check thrust bearing clearance / float	Per OEM spec, typ. 0.30-0.45 mm	☐	
4	Inspect labyrinth / carbon ring gland seals for wear	Replace if worn beyond limit	☐	
5	Record radial and axial clearances at all diaphragm stages	Clearance record sheet	☐	
6	Cross-check cold (assembly) clearance against expected hot clearance	Account for thermal growth	☐	

4 Governing & Trip System

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Function-test governor / speed control linkage	Free movement, no binding	☐	
2	Test overspeed trip mechanism (mechanical or electronic)	Trip at rated OSP setting	☐	
3	Inspect trip valve, servo motor, and emergency stop valve	No leakage on seat test	☐	
4	Check governor oil relay and control oil pressure	Per OEM setpoint	☐	

5 Reassembly & Dynamic Balancing

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Dynamic balance rotor assembly (if disturbed)	ISO 21940-11, Grade G2.5 typical	☐	



#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
2	Calculate permissible residual unbalance for the rotor mass/speed	$U_{per} = 9549 \times G \times M / n$ (g.mm)	☐	
3	Torque all coupling and casing bolts to spec (record values)	Use calibrated torque wrench	☐	
4	Re-check all running clearances after assembly	Match Section 3 records	☐	
5	Align turbine-driven equipment (laser or dial indicator)	Per machine alignment tolerance	☐	

Balance tolerance formula (ISO 21940-11 / ISO 1940-1): permissible residual unbalance U_{per} (g.mm) = $9549 \times G \times M / n$, where G is the balance grade in mm/s ($G2.5$ is standard for turbine and compressor rotors, $G6.3$ for general industrial rotors such as fans and pump impellers), M is rotor mass in kg, and n is maximum service speed in RPM. Example: a 200 kg rotor at 3,000 RPM balanced to $G2.5$ has $U_{per} = 9549 \times 2.5 \times 200 / 3,000 \approx 1,592$ g.mm total, split across the two correction planes.

6 Pre-Commissioning Sign-Off

#	CHECKPOINT	SPEC / REFERENCE	DONE	REMARKS
1	Lube oil flushing completed and cleanliness verified	ISO 4406 target class met	☐	
2	Barring gear operation checked, rotor rotates freely	No abnormal rubbing/noise	☐	
3	Instrumentation (RTDs, vibration probes, gauges) calibrated	Calibration certificates	☐	
4	Steam line blown through / warmed as per procedure	No debris on strainer screen	☐	
5	Final walk-down and housekeeping completed	Area clear of tools/debris	☐	

7 Post-Overhaul Vibration Acceptance (ISO 20816-3)

For steam turbines, ISO 20816-3 Group 1 (large machines, rated power above 300 kW, normally on sleeve/journal bearings) sets the RMS bearing-housing velocity zone boundaries below. A newly overhauled machine should run in Zone A or B; Zone C calls for investigation before continued service, and Zone D means stop and correct immediately.

ZONE BOUNDARY	RIGID FOUNDATION (mm/s RMS)	FLEXIBLE FOUNDATION (mm/s RMS)
A / B (new / good condition)	2.3	3.5
B / C (acceptable long-term)	4.5	7.1
C / D (unsatisfactory / damage risk)	7.1	11.0



Note: Land-based steam turbines and generators above 50 MW fall under ISO 20816-2, which uses similar principles but different limits. Confirm which part applies for very large units, and use shaft (proximity-probe) displacement limits instead of casing velocity where fluid-film bearings and permanently installed probes are fitted.

OK

Sign-Off

ENGINEER (KESHAV ENTERPRISES)		DATE	
CUSTOMER REPRESENTATIVE		DATE	

This checklist is a general field guide based on standard steam turbine overhaul practice. Always follow the OEM's specific manual, clearance charts, and torque values for your turbine make and model. Keshav Enterprises accepts no liability for outcomes arising from deviation from OEM instructions.